ENTERPRISE BANKING NETWORK INFRASTRUCTURE

Technical Design & Implementation Documentation

1. EXECUTIVE SUMMARY

This document details the architecture and security implementation for an Enterprise Banking Network Infrastructure. Designed to meet compliance, strict security, and high-availability criteria, this infrastructure integrates Cisco switching/routing, FortiGate Next-Generation Firewalls (NGFW), and Windows Server services. The resulting architecture ensures a highly segmented, resilient, and centrally managed environment for banking operations.

2. PROJECT OBJECTIVES

- Achieve maximum network uptime by removing single points of failure via device and path redundancy.
- Enforce rigorous traffic segmentation between banking operations, administrative tiers, and public internet networks.
- Implement centralized identity and access management for corporate users and devices.
- Provide robust edge security with dynamic multi-provider WAN connectivity and filtering policies.

3. NETWORK SEGMENTATION (VLAN DESIGN)

Logical network segmentation is established at Layer 2 using Virtual Local Area Networks (VLANs), restricting communication paths to explicitly authorized flows.

VLAN ID	Network Name	Operational Purpose & Access Control
VLAN 10	Management	Executive traffic, internal corporate administration, and management devices.
VLAN 20	Tellers	Front-line branch banking teller systems handling daily transactions. Restricted tier.
VLAN 30	Accounting & Finance	Back-office financial analysis, accounting ledger software, and payroll systems.
VLAN 40	IT Department	Systems engineering, network administration, and core infrastructure management access.
VLAN 50	ATM Network	Highly isolated segment dedicated exclusively to Automated Teller Machines (ATMs).
VLAN 60	Servers	Data center segment hosting Active Directory, DHCP pools, databases, and network services.

4. LAYER 2 SWITCHING & TOPOLOGY CONTROL

4.1 VLAN Distribution via VTP v3

VLAN Trunking Protocol (VTP) Version 3 is implemented for centralized VLAN management. VTP v3 prevents database configuration overrides by unverified network switches and ensures proper consistency across all active trunks.

4.2 Link Aggregation via PAgP

Backbone links connecting distribution devices utilize Cisco Port Aggregation Protocol (PAgP) to form logical EtherChannels.

- Interfaces are configured in PAgP Desirable Mode to actively negotiate bundle formation.
- This configuration aggregates physical link bandwidth and provides automatic sub-second failover if a physical link within the bundle degrades or drops.

4.3 Access-Layer Port Security

Cisco Port Security is applied to access switch ports to prevent unauthorized device access. Using the sticky MAC address feature, ports automatically lock and shut down if an unauthorized hardware address is detected.

5. LAYER 3 ROUTING & HIGH AVAILABILITY

First-hop default gateway redundancy is combined with standard interior dynamic routing protocols to maintain continuous data flow.

First-Hop Redundancy Parameters (VRRP)

Virtual Router Redundancy Protocol (VRRP) provides gateway redundancy across all production subnets:

- Distribution Switch 1 (Active Master): 192.168.X.1
- Distribution Switch 2 (Standby Backup): 192.168.X.2
- Virtual Redundant Gateway Address: 192.168.X.100 (Configured on all end-hosts)

5.1 Dynamic Routing Protocol (OSPF)

Open Shortest Path First (OSPF) is deployed as the Interior Gateway Protocol (IGP). Dynamic adjacencies are maintained between distribution switches, routers, and the edge FortiGate firewall to ensure fast topology convergence and automated path determination.

6. PERIMETER DEFENSE & FIREWALL SECURITY POLICIES

A FortiGate Next-Generation Firewall operates at the edge, enforcing administrative parameters between the corporate networks and the public internet.

6.1 Access Policies & Content Filtering

- **Inter-VLAN Enforcement:** Stateful security rules prevent cross-communication between unauthorized segments (e.g., blocking standard traffic from Teller VLAN 20 to Server VLAN 60).
- **Web Filtering:** Content restriction profiles govern outbound web traffic. Social media platforms, specifically Facebook, are completely blocked for users assigned to the Teller Department (VLAN 20).

6.2 Edge SD-WAN & WAN Failover

Reliable internet connectivity is established through software-defined WAN (SD-WAN) using dual providers:

1. Orange ISP: Primary enterprise internet data path.
2. WE ISP: Secondary circuit utilized for backup and automatic failover.

FortiGate SD-WAN policies continuously track link performance metrics. If the primary provider link goes down, active sessions instantly shift to the backup connection with minimal operational disruption.

7. CENTRALIZED NETWORK SERVICES (WINDOWS SERVER)

Core infrastructure identity authentication and addressing services are handled by a dedicated Windows Server instance.

- **Active Directory Domain Services (AD DS):** Implements centralized domain authentication. Logical Organizational Units (OUs) align with physical bank departments to ensure standardized security policy applications.
- **DHCP Services:** Centralized DHCP scopes manage dynamic network addressing, precisely mapped to each independent VLAN subnet to eliminate duplication or addressing conflicts.